



Name: _____

Class: _____

Date: _____

energy changes

Time:

Marks:

62 marks

Comments:

1.

This question is about displacement reactions.

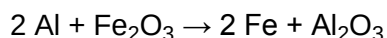
- (a) The displacement reaction between aluminium and iron oxide has a high activation energy.

What is meant by 'activation energy'?

(1)

- (b) A mixture contains 1.00 kg of aluminium and 3.00 kg of iron oxide.

The equation for the reaction is:



Show that aluminium is the limiting reactant.

Relative atomic masses (A_r): O = 16 Al = 27 Fe = 56

(4)

Magnesium displaces zinc from zinc sulfate solution.

- (c) Complete the ionic equation for the reaction.

You should include state symbols.



(2)

- (d) Explain why the reaction between magnesium atoms and zinc ions is both oxidation and reduction.

(2)

(Total 9 marks)

2.

This question is about citric acid ($\text{C}_6\text{H}_8\text{O}_7$).

Citric acid is a solid.

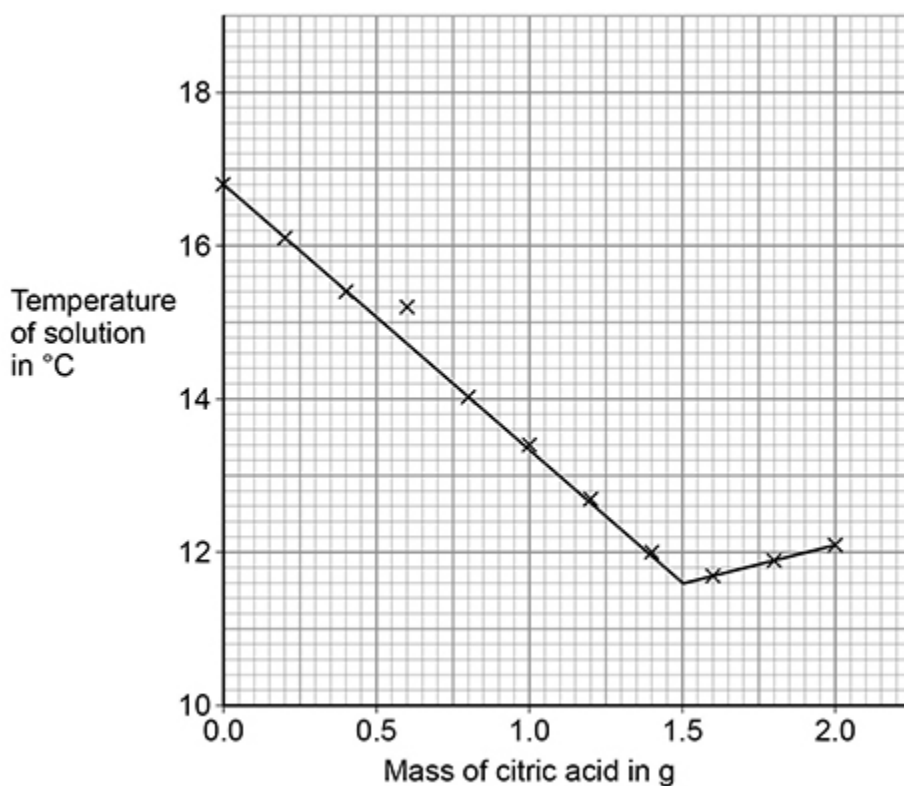
A student investigated the temperature change during the reaction between citric acid and sodium hydrogencarbonate solution.

This is the method used.

1. Pour 25 cm^3 of sodium hydrogencarbonate solution into a polystyrene cup.
2. Measure the temperature of the sodium hydrogencarbonate solution.
3. Add 0.20 g of citric acid to the polystyrene cup.
4. Stir the solution.
5. Measure the temperature of the solution.
6. Repeat steps 3 to 5 until a total of 2.00 g of citric acid has been added.

The student plotted the results on a graph.

The student's graph is shown below.



- (a) The graph shows an anomalous point when 0.60 g of citric acid was added. This was caused by the student making an error.

The student correctly:

- measured the mass of the citric acid
- read the thermometer
- plotted the point.

Suggest **one** reason for the anomalous point.

(1)

- (b) Explain the shape of the graph in terms of the energy transfers taking place.

You should use data from the graph above in your answer.

(3)

- (c) A second student repeated the investigation using a metal container instead of the polystyrene cup. The container and the cup were the same size and shape.

Sketch a line on above graph to show the second student's results until 1.00 g of citric acid had been added. The starting temperature of the solution was the same.

Explain your answer.

(3)

The student used a solution of citric acid to determine the concentration of a solution of sodium hydroxide by titration.

- (d) The student made 250 cm^3 of a solution of citric acid of concentration 0.0500 mol/dm^3

Calculate the mass of citric acid ($\text{C}_6\text{H}_8\text{O}_7$) required.

Relative atomic masses (A_r): $\text{H} = 1$ $\text{C} = 12$ $\text{O} = 16$

Mass = _____ g

(3)

This is part of the method the student used for the titration.

1. Measure 25.0 cm^3 of the sodium hydroxide solution into a conical flask using a pipette.
2. Add a few drops of indicator to the flask.
3. Fill a burette with citric acid solution.

- (e) Describe how the student would complete the titration.

(3)

- (f) Give **two** reasons why a burette is used for the citric acid solution.

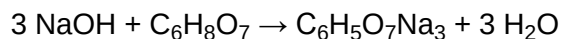
1 _____

2 _____

(2)

- (g) 13.3 cm³ of 0.0500 mol/dm³ citric acid solution was needed to neutralise 25.0 cm³ of sodium hydroxide solution.

The equation for the reaction is:



Calculate the concentration of the sodium hydroxide solution in mol/dm³

Concentration = _____ mol/dm³

(3)

(Total 18 marks)

3.

A student investigated the temperature change in the reaction between dilute sulfuric acid and potassium hydroxide solution.

This is the method used.

1. Measure 25.0 cm³ potassium hydroxide solution into a polystyrene cup.
2. Record the temperature of the solution.
3. Add 2.0 cm³ dilute sulfuric acid.
4. Stir the solution.
5. Record the temperature of the solution.
6. Repeat steps 3 to 5 until a total of 20.0 cm³ dilute sulfuric acid has been added.

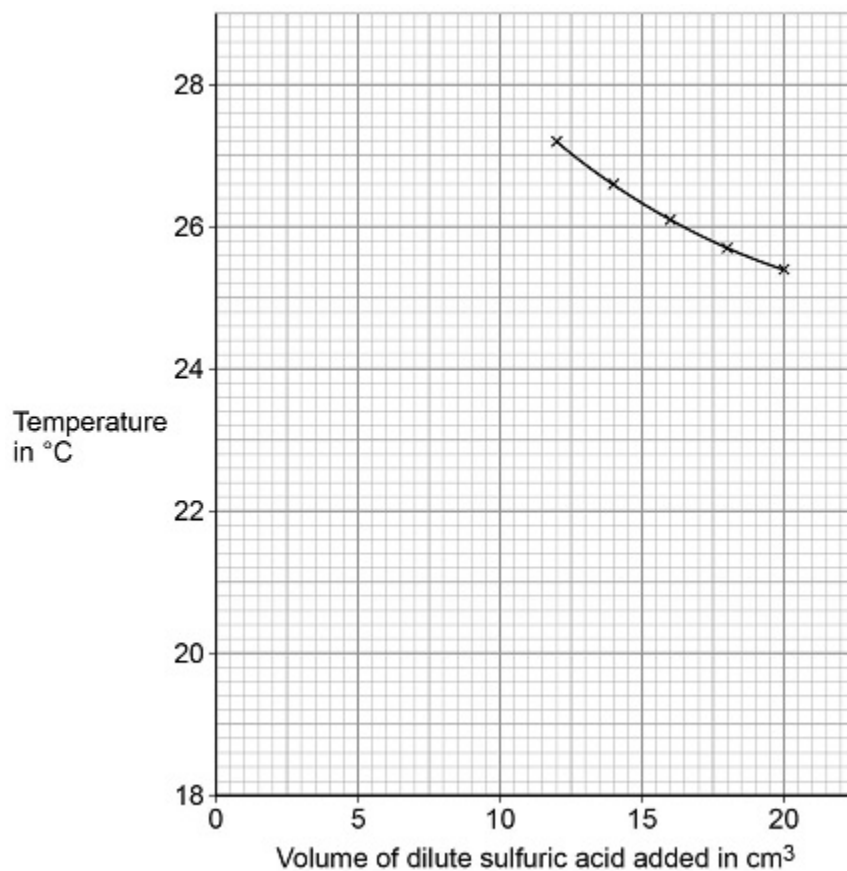
- (a) Suggest why the student used a polystyrene cup rather than a glass beaker for the reaction.

(2)

The following table shows some of the student's results.

Volume of dilute sulfuric acid added in cm ³	Temperature in °C
0.0	18.9
2.0	21.7
4.0	23.6
6.0	25.0
8.0	26.1
10.0	27.1

The figure below shows some of the data from the investigation.



(b) Complete the figure:

- plot the data from the table
- draw a line of best fit through these points
- extend the lines of best fit until they cross.

(4)

- (c) Determine the volume of dilute sulfuric acid needed to react completely with 25.0 cm³ of the potassium hydroxide solution.

Use the figure above.

Volume of dilute sulfuric acid to react completely = _____ cm³

(1)

- (d) Determine the overall temperature change when the reaction is complete.

Use the figure above.

Overall temperature change = _____ °C

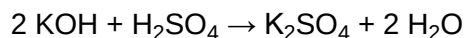
(1)

- (e) The student repeated the investigation.

The student used solutions that had different concentrations from the first investigation.

The student found that 15.5 cm³ of 0.500 mol/dm³ dilute sulfuric acid completely reacted with 25.0 cm³ of potassium hydroxide solution.

The equation for the reaction is:



Calculate the concentration of the potassium hydroxide solution in mol/dm^3 and in g/dm^3

Relative atomic masses (A_r): H = 1 O = 16 K = 39

Concentration in mol/dm^3 = _____ mol/dm^3

Concentration in g/dm^3 = _____ g/dm^3

(6)

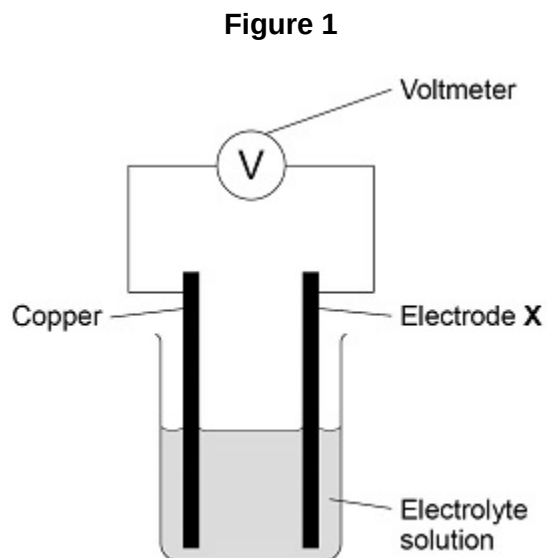
(Total 14 marks)

4.

This question is about chemical cells.

A student investigated the voltage produced by different chemical cells.

Figure 1 shows the apparatus.



This is the method used.

1. Use cobalt as electrode **X**.
2. Record the cell voltage.
3. Repeat steps 1 and 2 using different metals as electrode **X**.

(a) Suggest **two** control variables used in this investigation.

1 _____

2 _____

(2)

The following table shows the student's results.

Electrode X	Voltage of cell in volts
cobalt	+0.62
copper	0.00
magnesium	+2.71
nickel	+0.59
silver	-0.46
tin	+0.48

(b) Write the six metals used for electrode **X** in order of reactivity.

Use the table above.

Justify your order of reactivity.

Most reactive _____

Least reactive _____

Justification _____

(4)

- (c) Which of the following pairs of metals would produce the greatest voltage when used as the electrodes in the cell?

Use the table above.

Tick (✓) **one** box.

Magnesium and cobalt

☐

Magnesium and tin

☐

Nickel and cobalt

☐

Nickel and tin

☐

(1)

- (d) Hydrogen fuel cells can be used to power different forms of transport.

Some diesel trains are being converted to run on hydrogen fuel cells.

A newspaper article referred to the converted trains as the new 'steam trains'.

Suggest why.

(2)

(Total 9 marks)

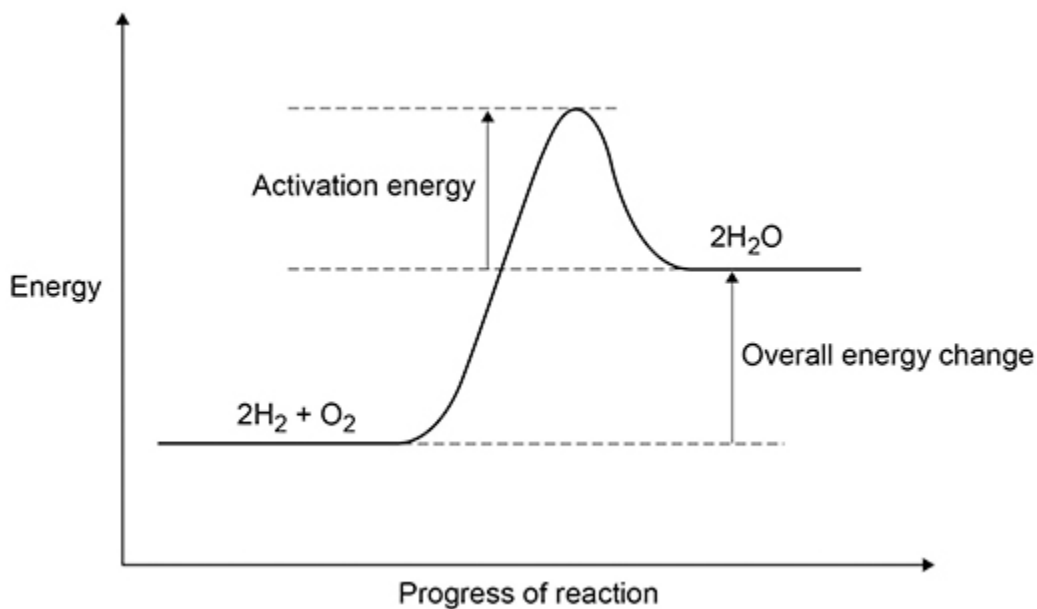
5.

The reaction between hydrogen and oxygen releases energy.

(a) A student drew a reaction profile for the reaction between hydrogen and oxygen.

Figure 1 shows the student's reaction profile.

Figure 1



The student made **two** errors when drawing the reaction profile.

Describe the **two** errors.

1 _____

2 _____

(2)

- (b) The reaction between hydrogen and oxygen in a hydrogen fuel cell is used to produce electricity.

Hydrogen fuel cells and rechargeable cells are used to power some cars.

Give **two** advantages of using hydrogen fuel cells instead of using rechargeable cells to power cars.

1 _____

2 _____

(2)

- (c) Reactions occur at the positive electrode and at the negative electrode in a hydrogen fuel cell.

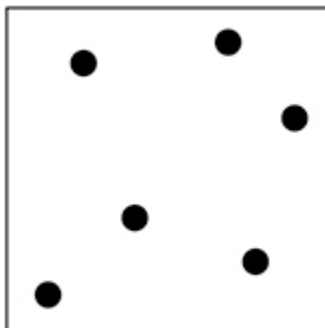
Write a half equation for **one** of these reactions.

(1)

- (d) The three states of matter can be represented by a simple particle model.

Figure 2 shows a simple particle model for hydrogen gas.

Figure 2



Give **two** limitations of this simple particle model for hydrogen gas.

1 _____

2 _____

(2)

- (e) The hydrogen gas needed to power a car for 400 km would occupy a large volume.

Suggest **one** way that this volume can be reduced.

(1)

- (f) The energy needed for a car powered by a hydrogen fuel cell to travel 100 km is 58 megajoules (MJ).

The energy released when 1 mole of hydrogen gas reacts with oxygen is 290 kJ

The volume of 1 mole of a gas at room temperature and pressure is 24 dm³

Calculate the volume of hydrogen gas at room temperature and pressure needed for the car to travel 100 km

Volume of hydrogen gas = _____ dm³

(4)

(Total 12 marks)

Mark schemes

1.

- (a) the (minimum) energy needed for particles to react
or
the (minimum) energy needed for a reaction to occur
allow the (minimum) energy needed to start a reaction

1

- (b) (M_r of Fe_2O_3 =) 160

1

(moles Fe_2O_3 = $\frac{3000}{160}$ =)
18.75 (mol)

allow correct use of incorrectly calculated M_r

1

(moles Al = $\frac{1000}{27}$ =) 37.0 (mol)

*allow 37.037037 (mol) correctly rounded to at least 2
significant figures*

*if both MP2 and MP3 are not awarded allow **1** mark for
0.01875 mol Fe_2O_3 **and** 0.037 mol Al*

1

(aluminium is limiting because)

37.0 mol is less than the (2×18.75 =) 37.5 mol (aluminium needed)

or

iron oxide is in excess because 18.75 mol is more than the ($\frac{37.0}{2}$ =) 18.5 mol (iron
oxide needed)

*allow correct use of incorrect number of moles from
steps 2 and/or 3*

alternative approaches:

approach 1:

(finding required mass of aluminium by moles method)

$$(M_r \text{ of Fe}_2\text{O}_3 =) 160 \text{ (1)}$$

$$(\text{moles Fe}_2\text{O}_3 = \frac{3000}{160} =)$$

$$18.75 \text{ (mol) (1)}$$

allow correct use of incorrectly calculated M_r

$$(\text{moles Al needed} = 18.75 \times 2 =) 37.5 \text{ (mol)}$$

and

$$(\text{mass Al needed} = 37.5 \times 27 =) 1012.5 \text{ (g) or } 1.0125 \text{ kg (1)}$$

allow correct use of incorrectly calculated moles of iron oxide

allow correct use of incorrectly calculated moles of aluminium needed

$$(\text{so) } 1.00 \text{ kg of aluminium is not enough (1)}$$

dependent on calculated mass of aluminium needed being greater than 1.00 (kg)

approach 2:

(finding required mass of aluminium by proportion method)

$$(M_r \text{ of Fe}_2\text{O}_3 =) 160 \text{ (1)}$$

$$(3.00 \text{ kg Fe}_2\text{O}_3 \text{ needs})$$

$$\frac{3.00}{160} \times 2 \times 27 \text{ (kg Al) (1)}$$

allow correct use of incorrectly calculated M_r

$$(=) 1.0125 \text{ (kg) (1)}$$

$$(\text{so) } 1.00 \text{ kg of aluminium is not enough (1)}$$

dependent on calculated mass of aluminium needed being greater than 1.00 (kg)

alternative approaches:

approach 3:
(finding required mass of iron oxide by moles method)

M_r of $\text{Fe}_2\text{O}_3 \Rightarrow 160$ (1)

(moles Al = $\frac{1000}{27} \Rightarrow 37.0$ (mol) (1)

allow 37.037037 (mol) correctly rounded to at least 2 significant figures

(moles Fe_2O_3 needed) = $\frac{37.0}{2} = 18.5$ (mol)

and

(mass Fe_2O_3 needed = $18.5 \times 160 \Rightarrow 2960$ (g) **or** 2.96 (kg) (1)

allow correct use of incorrectly calculated moles of aluminium

allow correct use of incorrectly calculated moles of iron oxide needed

allow correct use of incorrectly calculated M_r

(so) 3.00 kg of iron oxide is an excess (1)

dependent on calculated mass of iron oxide needed being less than 3.00 (kg)

approach 4:
(finding required mass of iron oxide by proportion method)

(M_r of $\text{Fe}_2\text{O}_3 \Rightarrow 160$ (1)

(1.00 kg Al needs) $\frac{1.00}{2 \times 27}$ (kg Fe_2O_3) (1)

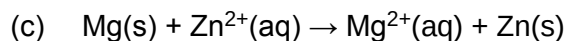
allow correct use of incorrectly calculated M_r

(=) 2.96 (kg) (1)

(so) 3.00 kg of iron oxide is an excess (1)

dependent on calculated mass of iron oxide needed being less than 3.00 (kg)

1



allow multiples

allow 1 mark for $\text{Mg}^{2+} + \text{Zn}$ with missing or incorrect state symbols

2

(d) magnesium (atoms) are oxidised because they lose electrons

1

(and) zinc (ions) are reduced because they gain electrons

if no other marks awarded allow 1 mark for magnesium (atoms) lose electrons and zinc (ions) gain electrons 1

1

[9]

2.

(a) didn't stir (the solution enough)

allow measured the temperature before the temperature stopped falling

allow measured the temperature too soon

1

(b) the temperature decreases (initially) because energy is taken in (by the reaction from the solution)

allow temperature decreases (initially) because the reaction is endothermic

when 1.5 g (of citric acid) is added the sodium hydrogencarbonate has all reacted

allow when the temperature reaches 11.6 °C the sodium hydrogencarbonate has all reacted

or

from 1.5 g the citric acid is in excess

allow after the temperature reaches 11.6 °C the citric acid is in excess

or

when 1.5 g (of citric acid) is added the reaction is complete

allow when the temperature reaches 11.6 °C the reaction is complete

(so) the temperature increases as energy is transferred from the room to the solution

allow (so) the temperature increases as energy is transferred from the excess citric acid to the solution

1

(c) less steep line starting at 16.8 °C **and** reaching 1.00 g (of citric acid)

ignore any part of the line drawn beyond 1.00 g

1

(as) metal is a better conductor

allow (as) polystyrene is a better insulator

1

(so) more energy is absorbed (from the surroundings)

allow (so) more heat is absorbed (from the surroundings)

1

(d) (M_r citric acid $\Rightarrow$) 192

$$(\text{moles} = \frac{250}{1000} \times 0.0500) = 0.0125$$

$$(\text{mass} = 0.0125 \times 192 \Rightarrow) 2.4 \text{ (g)}$$

1

*allow correct use of an incorrectly calculated M_r
allow correct use of an incorrectly calculated number of
moles*

1

alternative approach:

$$(\text{concentration} = 0.0500 \times 192)$$

$$= 9.6 \text{ (g/dm}^3\text{) (1)}$$

allow correct use of an incorrectly calculated M_r

$$(\text{mass} = \frac{250}{1000} \times 9.6 \Rightarrow) 2.4 \text{ (g) (1)}$$

*allow correct use of an incorrectly calculated
concentration in g/dm³*

(e) add the citric acid (to the flask) until there is a (permanent) colour change

ignore colours of indicator

1

measure / record the volume (of citric acid) added

allow take the final (and initial) burette reading

1

any **one** from:

- swirl
- use a white tile
- add the citric acid dropwise (near the end-point)
- repeat **and** calculate a mean

allow add the citric acid slowly (near the end-point)

1

(f) any **two** from:

- can add (the citric acid) in small increments
*allow can add (the citric acid) drop by drop
allow can add (the citric acid) slowly*
- can measure variable volumes
allow has a scale 2
- more accurate than a measuring cylinder

2

(g) (moles citric acid = $\frac{13.3}{1000} \times 0.0500$) = 0.000665

1

(moles NaOH = 3×0.000665) = 0.001995

allow correct use of an incorrectly calculated number of moles of citric acid

1

(conc = $\frac{1000}{25} \times 0.001995$) = 0.0798 (mol/dm³)

allow 0.08 or 0.080 (mol/dm³)

allow correct use of an incorrectly calculated number of moles of NaOH

1

alternative approach:

$$\frac{25.0 \times \text{conc NaOH}}{13.3 \times 0.0500} = \frac{3}{1} \quad (1)$$

allow $\frac{13.3 \times 0.0500}{25.0 \times \text{conc NaOH}} = \frac{1}{3}$

(conc NaOH =) $3 \times \frac{13.3 \times 0.0500}{25.0} \quad (1)$

= 0.0798 (mol/dm³) (1)

allow 0.08 or 0.080 (mol/dm³)

[18]

3.

(a) polystyrene is a better (thermal) insulator

allow polystyrene is a poorer (thermal) conductor

1

(so) reduces energy exchange (with the surroundings)

allow (so) reduces energy / heat loss (to the surroundings)

1

(b) all six points plotted correctly

allow a tolerance of $\pm \frac{1}{2}$ a small square

allow 1 mark for at least 3 points plotted correctly

2

line of best fit through points plotted from the table

1

both lines of best fit extrapolated correctly until they cross

(c) 11 (cm³)

allow ecf from part (b)

allow answers in the range 10.75 to 11.25 (cm³)

allow a tolerance of $\pm \frac{1}{2}$ a small square

1

(d) (27.5 – 18.9) = 8.6 (°C)

allow ecf from part (b)

allow answers in the range 8.5 to 8.7 (°C)

allow a tolerance of $\pm \frac{1}{2}$ a small square

1

(e)

*an answer of 0.62 (mol/dm³) for concentration in mol/dm³ scores **4** marks*

*an answer of 0.31 (mol/dm³) for concentration in mol/dm³ scores **3** marks*

$$(\text{moles H}_2\text{SO}_4 = 0.500 \times \frac{15.5}{1000}) = 0.00775$$

1

$$(\text{moles KOH} = 2 \times \text{moles H}_2\text{SO}_4 = 2 \times 0.00775) = 0.0155$$

allow correct calculation using incorrectly calculated value of moles of H₂SO₄

1

$$(\text{conc KOH} = \text{moles KOH} \times \frac{1000}{25.0}) = 0.0155 \times \frac{1000}{25.0}$$

allow correct calculation using incorrectly calculated value of moles of KOH

1

$$= 0.62 \text{ (mol/dm}^3\text{)}$$

allow correct answer using incorrectly calculated value of moles of KOH

1

$$(M_r \text{ KOH} =) 56$$

1

$$(\text{conc} = M_r \times \text{conc in mol/dm}^3 = 56 \times 0.62) = 34.7 \text{ (g/dm}^3\text{)}$$

allow 35 or 34.72 (g/dm³)

allow correct answer using incorrectly calculated value of concentration in mol/dm³ and/or incorrect M_r

1

alternative approach for step 1 to step 4

$$\frac{2}{1} = \frac{25 \times \text{conc KOH}}{15.5 \times 0.500} \quad (2)$$

$$(\text{conc KOH}) = \frac{2 \times 15.5 \times 0.500}{25.0} \quad (1)$$

$$= 0.62 \text{ (mol/dm}^3\text{)} \quad (1)$$

allow 1 mark if mole ratio is incorrect

1

[14]

4.

(a) any **two** from:

- temperature (of solution)
ignore room temperature
- concentration of electrolyte / solution
- compound / ions in electrolyte / solution
allow volume of electrolyte / solution
allow size of electrode
allow distance between electrodes
do not accept electrode X unqualified
do not accept (measured) voltage

2

(b) order:

(most reactive) magnesium

cobalt

nickel

tin

copper

(least reactive) silver

allow 1 mark for magnesium, cobalt, nickel, tin in order at top

allow 1 mark for copper and silver in order at the bottom

2

justification:

the higher the (positive) voltage, the more reactive (the metal)

allow the most reactive (metal) has the highest voltage

1

silver has a negative voltage because silver is less reactive than copper

1

(c) magnesium and tin

1

(d) (in a fuel cell) hydrogen is oxidised (to produce water)
allow (in a fuel cell) hydrogen reacts with oxygen (to produce water)

1

water is produced / released as gas / vapour / steam

if no other mark awarded, allow 1 mark for fuel cells produce water

1

[9]

5.

(a) the activation energy should be from the reactants (line to the peak)
ignore description of where the activation energy is on the diagram

1

the products (line) should be below the reactants (line)

or

the products should have less energy than the reactants

allow the product (line) is above the reactants (line)

allow the products have more energy than the reactants

allow the profile shows an endothermic reaction

ignore the arrow for the overall energy change should

point downwards

1

(b) any **two** from: (hydrogen fuel cells)

allow converse arguments for a rechargeable cell

- no toxic chemicals to dispose of at the end of the cell's life
- take less time to refuel (than to recharge rechargeable cells)
- travel further before refuelling (than before recharging rechargeable cells)
allow has a greater range
- no loss of efficiency (over time)
allow does not lose capacity / range in cold weather

2

(c) any **one** from:

allow multiples

- $\text{H}_2 \rightarrow 2 \text{H}^+ + 2 \text{e}^-$
allow $\text{H}_2 - 2 \text{e}^- \rightarrow 2 \text{H}^+$
- $\text{O}_2 + 4 \text{H}^+ + 4 \text{e}^- \rightarrow 2 \text{H}_2\text{O}$
allow $\text{H}_2 + 2 \text{OH}^- - 2 \text{e}^- \rightarrow 2 \text{H}_2\text{O}$
- $\text{H}_2 + 2 \text{OH}^- \rightarrow 2 \text{H}_2\text{O} + 2 \text{e}^-$
- $\text{O}_2 + 2 \text{H}_2\text{O} + 4 \text{e}^- \rightarrow 4 \text{OH}^-$

1

(d) any **two** from:

- hydrogen is not shown as H_2 / molecules
- particles are shown as spheres
- particles are shown as solid
- does not show the (weak) forces (between particles)
- does not show the movement / speed (of particles)
- is only two-dimensional

2

(e) any **one** from:

- under (higher) pressure
allow increase concentration
- cool
allow condense
- absorb / adsorb in a solid
allow store as a liquid / solid
allow develop more efficient engines

1

(f) (58 MJ $\Rightarrow$) 58 000 kJ

or

(290 kJ $\Rightarrow$) 0.290 MJ

allow (58 MJ $\Rightarrow$) 58 000 000 J

and

(290 kJ $\Rightarrow$) 290 000 J

1

(moles = $\frac{58000}{290}$ **or** $\frac{58}{0.290}$)

*allow correct use of an incorrectly converted or
unconverted value of energy*

1

(volume $\Rightarrow$) 200×24

*allow correct use of an incorrectly calculated number of
moles of hydrogen*

1

= 4800 (dm³)

1

alternative approach:

(58 MJ $\Rightarrow$) 58 000 kJ (1)

(energy released per dm³ = $\frac{290}{24}$ $\Rightarrow$) 12.08333 (kJ/dm³) (1)

(volume $\Rightarrow$) $\frac{58000}{12.08333}$ (1)

*allow correct use of an incorrectly converted or
unconverted value of energy*

*allow correct use of an incorrectly calculated energy
released per dm³*

= 4800 (dm³) (1)

[12]

Examiner reports

1.

- (a) Most students were able to express the idea that activation energy is the minimum energy required to start a reaction. One common error was to say that it is the energy needed to make particles collide.
- (b) A variety of approaches was seen. The majority of students calculated the number of moles of both aluminium and of iron oxide and then attempted with varying degrees of success to explain why the number of moles of aluminium was insufficient for the iron oxide. Common errors were to omit to convert the mass from kilograms to grams, and/or using 54 as the ' M_r ' of aluminium. Of those who did correctly calculate the number of moles of each reactant, some had difficulty in expressing clearly the idea of 2 moles of aluminium being needed to react with 1 mole of iron oxide, and therefore 37.0 moles of aluminium is less than the 37.5 moles required, or that 18.75 moles of iron oxide is more than the 18.5 moles required. Other students started with the mass of one reagent and calculated the required mass of the other, either by the moles method or by proportion. These students found it easier to express why there was insufficient aluminium or excess iron oxide.
- (c) Few students scored both marks. Aqueous zinc was often seen as a response. Many students did not understand the idea of an ionic equation and tried to include the sulfate ion.
- (d) Only a third of students scored both marks, but a lot of students were confused about which species gained electrons and which lost them. Definitions of oxidation and reduction in terms of gain and loss of oxygen were not appropriate here and were not credited.

2.

- (a) A third of students scored this mark; some realised that the most likely error was lack of stirring; fewer suggested that the temperature was measured too early, before the temperature had stopped falling. Many suggestions contradicted the information given.
- (b) Students had great difficulty in expressing their answers using correct terminology, often using energy and temperature interchangeably. They generally find endothermic reactions difficult; they do not seem to realise that the energy taken in is not staying as thermal energy but is being converted to chemical energy. Some students gained the mark for saying that the temperature initially falls because energy is being taken in, showing an endothermic reaction. A few noted that when 1.5 g of citric acid had been added the reaction had finished and that the citric acid was in excess from there on. However, it was very rare for a correct explanation for the subsequent rise in temperature to be given. Some thought the reaction suddenly became exothermic, or that it suddenly reversed (showing a lack of understanding of reversible reactions).
- (c) Many students realised that the difference was due to the better thermal conductivity of the metal. However, many of those merely referred to the metal as a good conductor, without comparing the metal to polystyrene. The majority of students drew a steeper line, stating that the metal would lose heat more quickly. Of those who drew the correct shallower line, few could explain that the metal would conduct thermal energy in from the surroundings more quickly, reducing the temperature drop in this endothermic reaction.
- (d) Many students were able to score all three marks. Common errors were to confuse concentration with number of moles and to fail to convert 250 cm^3 to 0.25 dm^3 correctly. The symbol 'm' for number of moles was commonly seen instead of 'n' – students consequently became confused between mass and number of moles. Students who set out their working step by step were able to gain partial credit, but those who simply had a jumble of numbers, with no explanation as to what the numbers represented, were not.
- (e) Most students scored at least two marks. Some students referred to 'amount' of acid added rather than volume, and many failed to take any burette readings at all. It was not uncommon to see a conical flask placed 'under' a white tile rather than on one.
- (f) This was not well answered. Most students who scored a mark did so for stating that the acid could be added drop by drop, or that the rate of flow could be controlled. Few stated that the burette could measure variable volumes (unlike a pipette). Many stated that a burette was 'more accurate' but did not say what they were comparing it to. Of those who did, most said it was more accurate than a pipette, which is not true. Some students confused a volumetric pipette with a dropping pipette.
- (g) This titration calculation was well done by many students. To gain full credit, working needed to be correct as well as the final answer. Students therefore need to show their working step by step in a way that can be followed by the examiner. Common errors were to omit to convert volumes to dm^3 , to attempt to do so but incorrectly, or to omit the mole ratio of citric acid to sodium hydroxide. As with question 09.4, students who incorrectly used the symbol 'm' to mean moles often confused themselves; they should be encouraged to use 'm' only for mass, and 'n' for number of moles.

3.

- (a) Almost a half of students scored 0 marks.

The mark scheme required the use of comparatives.

Common correct responses were:

- Better insulator
- Less heat loss

Common responses seen that were not credited included:

- good insulator
- no heat loss.

- (b) This graph question was well answered with approximately half of students scoring full marks.

Some students attempted to draw a straight line of best fit. In most cases, this was inappropriate.

Some students did not draw their line so that the two lines of best fit crossed. Instead, they drew a curved peak.

- (c) Approximately 60% of students were able to determine the neutralisation point from their graph.

In some cases, students misunderstood the scale of the x-axis, for example, reading 11 cm³ as 10.1 cm³.

- (d) Only approximately a third of students correctly determined the overall temperature change for the reaction from their graph.

A common incorrect answer was 8.2 °C (from subtraction of the highest and lowest temperature values in the table).

- (e) Students found this titration calculation difficult and approximately a third of students scored 0 marks.

The marks for calculating the number of moles of sulfuric acid and for calculating the relative formula mass of potassium hydroxide were the most scored. Even when these values were incorrect, this did not prevent students from gaining marks in subsequent steps.

The correct answer for the concentration in mol/dm³ was achieved more often than the subsequent conversion to g/dm³. A common error was to use a conversion factor of 112 ($2 \times M_r(\text{KOH})$) rather than 56.

4.

- (a) Students found this question difficult and were quite often imprecise in their description of the control variables. The use of the word amount is unlikely to gain credit and students should be encouraged to use scientific terminology such as 'volume'.

The most common control variables given were volume of solution and concentration of solution.

Common responses seen that were not credited included:

- amount of solution
- the electrolyte
- the copper electrode.

- (b) Most students scored the 2 marks for the order of reactivity of the top four metals and the justification for this order.

The most common error was then to put silver above copper in the order of reactivity.

However, only a small proportion of students were able to explain the negative voltage for silver in terms of its reactivity compared with copper.

- (c) Approximately half of students correctly identified 'magnesium and tin' as the pair of metals to produce the greatest voltage.

'Magnesium and cobalt' was a strong distractor, presumably because it also contained the most reactive metal.

- (d) Only a small proportion of students scored both marks. The mark for describing the reaction that produces water in a hydrogen fuel cell was the least frequently scored.

Often, students gained the compensation mark for the idea that hydrogen fuel cells produce water.

However a significant number of students gave incomplete responses indicating that hydrogen fuel cells produce steam (with no reference to water).

Some common misconceptions seen were:

- Hydrogen fuel cells release hydrogen gas.
- Hydrogen burns in a hydrogen fuel cell.

5.

- (a) Many students spotted that the curve was for an endothermic reaction rather than exothermic. Many also realised the activation energy was incorrect but could not always express correctly that activation energy should start at the energy of the reactants rather than the products.
- (b) Most students did not score well on this question. There were many vague references to 'better for the environment' without any justification for that statement. Many students referenced the fact that water is the only product, failing to realise that rechargeable cells produce no waste products at all. Others failed to realise that the infrastructure for recharging cells is actually better than that for refuelling with hydrogen. Some students were effectively giving advantages of hydrogen fuel cells over petrol engines, instead of answering the question. Others failed to express themselves clearly; saying that hydrogen fuel cells take less time to charge which did not gain credit, as the cells are refuelled, not recharged.
- (c) Many students did not realise that either hydrogen or oxygen had to be a reactant. Many impossible species such as e^{2-} or O_2^+ were seen.
- (d) Many students were able to score at least one mark but only one in six scored both. Many noted that the movement of the particles was not shown; many also noted that the particles are not shown as diatomic, or that the forces between them are not shown.
- (e) This was quite well answered, with most students favouring compressing the gas rather than cooling to condense the gas. A number of students incorrectly stated that hydrogen can be taken in from the atmosphere.
- (f) A third of students were able to score full marks. It was not possible to give partial credit for working when the students did not explain what they were doing; too often there was a jumble of numbers with various operators between them, with no words or units. No credit was given for multiplying a value by 24 unless that value was identified as a number of moles.